Gender Bias in Profession Classification

A Counterfactual Evaluation of DistilBERT Using the Bias in Bios Dataset

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This project investigates gender bias in a DistilBERT-based profession classifier trained on the Bias in Bios dataset. Using a counterfactual evaluation methodology, it measures how swapping gendered pronouns and markers in professional biographies changes model predictions and confidence scores — across 28 professions and two categories (STEM vs. Non-STEM). The pipeline is fully modular: data preparation, model inference, metric computation, and visualization are separated into independent modules. Key findings reveal that profession predictions are sensitive to gender cues even when professional content is held constant, with differential bias patterns across STEM and Non-STEM fields.

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1. Background and Motivation

Profession classification models trained on biographical text have been shown to encode and amplify societal gender stereotypes. When these models associate professional roles with specific genders — surgeon with male, nurse with female — they risk reinforcing discriminatory patterns in downstream applications such as resume screening, hiring recommendation, and career guidance systems.

The Bias in Bios dataset provides a structured benchmark for studying this problem: it contains real professional biographies scraped from the web, labeled by profession and ground-truth gender. This project uses counterfactual data augmentation — systematically swapping gendered pronouns and markers — to isolate how much of a classifier's prediction is driven by professional content versus gender cues.

This question is directly relevant to responsible AI deployment. A classifier that changes its profession prediction when only gender markers change is not measuring professional expertise — it is measuring gender stereotypes encoded in training data.

2. Research Question and Contributions

Core Research Question

To what extent does a DistilBERT-based profession classifier rely on gendered language cues rather than professional content — and does this reliance differ between STEM and Non-STEM professions?

Research Contributions

- Counterfactual evaluation pipeline for measuring gender bias in profession classifiers
- Flip rate and confidence shift metrics disaggregated across 28 professions
- STEM vs. Non-STEM bias comparison using a modular, reproducible Python codebase
- Visualization suite: flip rate bar charts, confidence histograms, violin plots, heatmaps
- Modular architecture separating data, model, analysis, and plotting concerns
- A replicable framework adaptable to other demographic bias axes (race, age, etc.)

3. Dataset and Counterfactual Methodology

3.1 Bias in Bios Dataset

The Bias in Bios dataset contains short professional biographies scraped from the web, labeled with profession (28 categories) and binary gender. It was introduced by De-Arteaga et al. (2019) specifically to study the relationship between gender representation and occupation classification accuracy in NLP models.

3.2 Counterfactual Data Augmentation

For each biography in the evaluation set, a counterfactual version is constructed by swapping gendered pronouns and markers (he/she, his/her, him/her, and name-derived gender signals) while leaving all professional content unchanged. This produces matched pairs: an original bio and a gender-swapped bio representing the same professional described with opposite gender markers.

This methodology controls for professional content by construction — any change in model prediction between the pair can be attributed to gender cues alone, not to differences in described expertise, credentials, or experience.

4. System Architecture and Pipeline

The project is structured as a modular Python pipeline. Each stage has a dedicated module with a single responsibility, enabling independent testing, replacement, and extension.

Stage	Module	Output
Data Preparation	data_utils.py	bios_pairs.csv — original + gender-swapped bio pairs
Model Loading	model_utils.py	Fine-tuned DistilBERT profession classifier
Counterfactual Inference	bias_analysis.py	Predictions + confidence scores for both bio versions
Metric Computation	analysis_utils.py	Flip rate, confidence shift (overall + by profession)
Visualization	plots.py	Flip rate bar, confidence histogram, violin, heatmap

4.1 Model

A DistilBERT model (`distilbert-base-uncased`) is fine-tuned on the Bias in Bios training split for 28-class profession classification. DistilBERT is selected for its computational efficiency and strong performance on text classification tasks, making it representative of the models commonly deployed in real-world profession classification pipelines.

4.2 Inference

Predictions are generated in batches of 16, with inputs truncated to 256 tokens. For each biography, the model outputs a predicted profession label and a confidence score (max softmax probability). Both original and counterfactual bios are processed identically, producing directly comparable output pairs.

5. Evaluation Metrics

Metric	Definition	Research Relevance
Flip Rate	% of bios where gender swap changes predicted profession	Measures classifier sensitivity to gendered language cues
Confidence Shift	Absolute change in max softmax probability after swap	Quantifies uncertainty introduced by gender marker changes
STEM vs. Non-STEM Breakdown	Flip rate and confidence shift by profession category	Tests whether bias concentrates in stereotypically gendered fields
Per-Profession Flip Rate	Flip rate disaggregated across 28 profession labels	Identifies which professions are most vulnerable to gender bias

6. Key Findings and Visualizations

6.1 Flip Rate

The flip rate measures the proportion of bio pairs where the model's predicted profession changes after gender swapping. A non-zero flip rate indicates that the classifier's decisions are not fully grounded in professional content — the model is using gender markers as predictive features for profession.

6.2 Per-Profession Patterns

Flip rates vary substantially across professions. Professions with strong gender stereotypes in training data — such as nurse, surgeon, and engineer — tend to show higher flip rates, reflecting the model's reliance on gender-profession co-occurrence patterns rather than content.

6.3 STEM vs. Non-STEM Differential

The STEM category (accountant, dentist, engineer, financial analyst, physician, professor, researcher, software developer, surgeon) is analyzed separately from Non-STEM professions. The violin plot and heatmap visualizations test whether bias is more pronounced in fields with well-documented gender representation gaps — a hypothesis consistent with the literature on occupational stereotyping in NLP.

6.4 Confidence Shift Distribution

The confidence shift histogram reveals how much the model's certainty changes when gender markers are swapped. High confidence shifts in the absence of prediction flips indicate that the model is internally uncertain even when it arrives at the same label — a subtler form of gender sensitivity that flip rate alone would miss.

7. Skill Development

- Fine-tuning transformer models (DistilBERT) for multi-class text classification
- Counterfactual data augmentation design and implementation
- Batch inference with GPU/CPU device management using PyTorch
- Bias metric design: flip rate, confidence shift, disaggregated analysis
- Research visualization: seaborn violin plots, heatmaps, histograms
- Modular Python codebase design for reproducible research pipelines

8. Limitations and Ethical Considerations

Methodological Constraints

- Binary gender framing — the dataset and swap methodology treat gender as binary, which does not reflect real gender diversity
- Pronoun-level swap may miss subtler gendered language patterns
- Evaluation on 1,000 sampled pairs; full-dataset results may differ

Scope and Generalizability

- Profession labels are culturally and geographically specific
- Model findings reflect the Bias in Bios dataset, not all professional contexts
- Flip rate measures sensitivity, not harm — deployment context determines impact

9. Reflection

The most technically instructive aspect of this project was designing metrics that distinguish between two different forms of gender sensitivity: prediction flips (the model changes its answer) and confidence shifts (the model becomes more uncertain without changing its answer). The second form is harder to detect and arguably more important for understanding how bias is encoded, since it suggests the model is relying on gender cues even when they do not ultimately determine the output.

The STEM/Non-STEM disaggregation was motivated by prior literature showing that gender representation gaps are particularly pronounced in technical fields. Testing this hypothesis computationally — rather than assuming it — is a small example of the kind of empirical rigor that distinguishes research from observation.

10. Conclusion and Future Scope

This project demonstrates that profession classifiers trained on biographical text encode gender bias that is measurable through counterfactual evaluation. The modular pipeline provides a reproducible framework for studying this bias across models, datasets, and demographic axes.

Future Directions

- Non-binary and gender-nonconforming counterfactual generation
- Extension to racial and ethnic bias using intersectional analysis
- Comparison across multiple model architectures (BERT, RoBERTa, GPT-2)
- Debiasing intervention experiments (adversarial training, data re-weighting)
- Application to real hiring or recommendation system audit contexts
- Longitudinal study of bias across model generations and training data vintages

11. Jury Summary

This project quantifies gender bias in a DistilBERT profession classifier using counterfactual data augmentation on the Bias in Bios dataset. By swapping gendered pronouns in professional biographies and measuring prediction changes, it produces two core metrics — flip rate and confidence shift — disaggregated across 28 professions and STEM/Non-STEM categories. The codebase is fully modular and reproducible. Key strength: the combination of quantitative bias metrics with disaggregated analysis and a visualization suite that makes findings interpretable. This is empirical AI fairness research, not speculation.

12. References

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